

Problem (1)

$$(a) \quad P[E=4] = \frac{5}{6}$$

$$\textcircled{b} \quad P[D=2, F=3] = \frac{3}{6} \cdot \frac{5}{6} = \frac{15}{36} = \frac{5}{12}$$

Problem (2)

(a) $E[D] = 2 \cdot \frac{1}{2} + 5 \cdot \frac{1}{2} = 3.5$

$$\bullet E[E] = 1 \cdot \frac{1}{6} + 4 \cdot \frac{5}{6} = 3.5$$

$$\bullet E[F] = 3 \cdot \frac{5}{6} + 6 \cdot \frac{1}{6} = 3.5$$

(b) Table of outcomes:

Diego (D)

[illegible]

(2/4)

By counting outcomes on the table,
Diego wins 21 times out of 36
equally likely outcomes. So

$$P[\text{Diego wins}] = \frac{21}{36} = \frac{7}{12} \approx .58$$

(C) First, we enumerate a table of outcomes:

		Diego (D)				
		2	2	2	5	5
Max (F)	3	Diego Loses (15 outcomes)			Diego Wins (15 outcomes)	
	3					
	3					
	3					
	3					
	6	Diego Loses (3 outcomes)			Diego Loses (3 outcomes)	

$$\therefore P[\text{Diego Wins}] = \frac{15}{36} = \frac{5}{12} \approx .42$$

(d) First, make a table:

		Diego ($\bar{E}$)					
		1	4	4	4	4	4
Max (F)	3	Diego Loses (5 out- comes)	Diego Wins (25 outcomes)				
	3						
	3						
	3						
	3						
	6	Diego Loses (6 outcomes)					

$$\therefore P[\text{Diego Wins}] = \frac{25}{36} \approx .69$$

Fun observation: dice E 'beats' dice F ,
 dice F 'beats' dice D ,
~~die~~ BUT dice D 'beats' dice E

like { $E = \text{Rock}$
 $F = \text{Paper}$
 $D = \text{Scissors}$ }

Problem (3)

Let's first make a table of dice sums:

		Red dice					
		1	2	3	4	5	6
Blue	1	2	3	4	5	6	7
	2	3	4	5	6	7	8
	3	4	5	6	7	8	9
	4	5	6	7	8	9	10
	5	6	7	8	9	10	11
	6	7	8	9	10	11	12

← Outcomes corresponding to B

← Outcomes corresponding to A.

By inspection of this table we have:

- $P[A] = \frac{6}{36} = \frac{1}{6}$
- $P[B] = \frac{6}{36} = \frac{1}{6}$
- $P[C] = \frac{3}{6} = \frac{1}{2}$
- $P[A \cup B] = P[A] + P[B] = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$

↑
(since $A \cap B = \emptyset$)

$$P[A \cap B^c] = P[A] = \frac{1}{6}$$

↑
since $A \cap B^c = A$